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The Polarization Paradox: How Suppressing Backfire Cascades Entrenches Ideological Segregation under Fragile Trust

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Abstract: Ideological polarization on modern social media platforms threatens democratic processes and platform economics through user churn and collapsing ad revenue. Platforms often respond with bridging-style feed interventions that suppress confrontational, cross-cutting interactions in the user's latitude of rejection. Using the Dynamic-BABE (Biased Assimilation & Behavioral Entrenchment) agent-based model, we study the long-term consequences of these interventions in a co-evolving network of interpersonal trust. A 2×2 factorial design crosses bridging (ON/OFF) with dyadic trust co-evolution (ON/OFF) across 30 replications per cell ($N = 500$, 200 steps). We find a Polarization Paradox: bridging sharply reduces cumulative churn from 39.18% to 0.38% ($d = 23.97$, $p < 0.001$) and raises step-wise revenue from \$121.64 to \$199.24 ($d = -23.97$, $p < 0.001$), yet final polarization rises from 0.9550 to 0.9904 ($d = -1.28$, $p < 0.001$). Separately, enabling trust co-evolution alone (Trust Only vs. Baseline) produces severe trust segregation (35.61 ± 31.50; median $= 28.58$, IQR $= [19.06, 40.61]$; $d = -1.55$, $p < 0.001$), with outgroup trust collapsing while ingroup trust concentrates in homogeneous silos. Seed-paired Bridge $\times$ Trust interaction tests show that bridging's retention benefits are robust to trust dynamics, while relational metrics exhibit modest moderation (pairwise Full Model vs. Trust Only contrasts for outgroup trust and segregation are non-significant after Bonferroni correction; the segregation interaction itself is non-significant under Wilcoxon). Overall, short-term retention gains need not improve long-term cohesion when disagreement is suppressed and trust is fragile.

Keywords: agent-based modeling; opinion dynamics; polarization; dyadic trust; algorithmic curation; social media

Introduction

1.1 The contemporary public square is no longer defined by physical geography; instead, digital platforms, feeds, and screens now stage the global public square. While these virtual arenas promised democratic liberation and social connection, they have instead spawned severe ideological and affective polarization (Bail et al. 2018; Levy 2021). Algorithms amplify disagreement in these online battlegrounds, fracturing the public square into mutually hostile echo chambers and systemic distrust.

1.2 For social media firms, polarization is not only a societal concern; it threatens platform economics. Hostile feeds inflict cognitive fatigue and emotional distress. Users respond by walking away—a response formalized by Hirschman (1970) as the classic "exit" option. As churn accelerates, active user bases and ad impressions shrink. Platforms also face advertiser sensitivity to reputational risk on volatile feeds.

In this paper, the "brand safety penalty" is a stylized revenue assumption used to model that pressure, not an empirically calibrated estimate from any single field study. Related work on feed ranking and political attitudes (Guess et al. 2023; Levy 2021) motivates attention to algorithmic curation, but does not directly measure brand-safety multipliers.

1.3 Platforms turn to algorithmic curation to stem exit. Beyond crude bans or post-hoc deletions, networks experiment with structural feed interventions that reduce exposure to confrontational, cross-cutting content before hostile exchanges escalate. We model such interventions as a probabilistic "bridging" filter that intercepts interactions in the psychological "latitude of rejection" (Sherif & Hovland 1961)—ideas so alien that exposure triggers defensive backlash rather than assimilation. Empirical feed experiments (Guess et al. 2023; Bail et al. 2018) show that algorithmic exposure regimes shape attitudes and behavior; our stylized bridge is an abstraction of suppression-oriented curation rather than a claim that any named commercial product implements this exact rule.

1.4 Yet suppression can create a counter-intuitive deadlock: the Polarization Paradox. Bridging may succeed in the short term—reducing churn, frustration, and revenue loss—while failing to reduce, and sometimes increasing, systemic polarization. By sanitizing feeds of cross-cutting friction, platforms can starve the network of contact needed to build mutual tolerance and relational trust.

1.5 We formalize this tension through the Dynamic-BABE (Biased Assimilation & Behavioral Entrenchment) model, an agent-based framework that extends classical opinion dynamics—including DeGroot (DeGroot 1974) and Hegselmann–Krause bounded confidence (Hegselmann & Krause 2002)—by combining Chen et al.'s entrenchment-based biased assimilation (Chen et al. 2021) with Social Judgment Theory (Sherif & Hovland 1961; Jager & Amblard 2005) and a co-evolving network of interpersonal trust. Trust is not a static parameter; it is a dyadic bond that co-evolves with conversation. Following Slovic (1993), trust accumulates slowly through agreement and erodes quickly under conflict.

1.6 Through a 2×2 factorial experiment crossing algorithm activation with dynamic trust, we show that sanitizing feeds can deprive agents of opportunities to cultivate mutual tolerance. Bridging dampens immediate conflict and suppresses frustration, but also reduces the low-friction interactions that could repair trust across divides. Relations left unmaintained wither under passive decay. Separately, enabling trust co-evolution produces trust segregation—trust pools inside homogeneous silos while outgroup trust collapses—independent of the bridging contrast that defines the Polarization Paradox.

1.7 Our contributions are threefold. First, we introduce Dynamic-BABE as an agent-based framework linking cognitive psychology, relational network dynamics, and stylized platform economics. Second, we demonstrate the Polarization Paradox: a trade-off between short-term retention and long-term cohesion under suppression. Third, we argue that purely avoidance-based moderation is incomplete if the goal is relational trust and mutual tolerance.

1.8 Replication materials are available at `https://github.com/TheCoolSam/BABEModel`; we intend to deposit the model on CoMSES Net to support permanent archival.[1]

Related Work

2.1 No model stands alone; its strength derives entirely from the theoretical bedrock anchoring it. To ground the Dynamic-BABE framework in established social science and computational methodologies, we dissect the literature across five domains: classical opinion dynamics, cognitive resistance, Social Judgment Theory, co-evolving relational trust, and algorithmic platform interventions.

Classical Models of Opinion Dynamics

2.2 Mathematical formalizations of social influence and consensus began with DeGroot's linear averaging framework (DeGroot 1974). In DeGroot's world, agents update their opinions by repeatedly computing weighted linear averages of their neighbors' beliefs. It is an elegant formulation. Yet, it suffers from a fatal limitation: in any connected network, it inevitably collapses into global consensus, failing to explain the stubborn disagreement and deep polarization that characterize real-world societies.

2.3 Seeking to bypass this consensus trap, Hegselmann & Krause (2002) introduced "bounded confidence," wherein agents influence each other only if their opinions fall within a pre-defined cognitive threshold. Deffuant et al. (2000) pursued a similar path in pairwise interactions, demonstrating that narrow confidence intervals fracture populations into isolated, non-communicating subgroups. These frameworks successfully prevent global uniformity. For a comprehensive overview of the mathematical challenges in traditional opinion dynamics, see Flache et al. (2017). However, they rely on a blunt, binary gatekeeper: outgroup agents are simply ignored. They fail to capture the active repulsion and aggressive cognitive resistance that fuel real-world toxic polarization.

Cognitive Resistance and Biased Assimilation

2.4 A psychologically realistic model must embrace the cognitive biases that govern how humans filter conflicting data. Here, the core mechanism is "biased assimilation," which Lord et al. (1979) famously demonstrated by exposing subjects to identical, mixed evidence on controversial topics. Their subjects did not synthesize the facts objectively; instead, they greedily devoured evidence that confirmed their priors while subjecting dissenting information to hostile, hyper-critical scrutiny. Balanced exposure thus backfires: rather than prompting convergence, it drives prior attitudes further apart.

2.5 Computational sociologists have long struggled to formalize this cognitive friction in agent-based networks. Dandekar et al. (2013) injected biased assimilation into a DeGroot-style model, proving that cognitive bias paired with network homophily accelerates polarization, pulling opinions to extreme, opposing poles. In a major leap forward, Chen et al. (2021) built the Biased Assimilation and Behavioral Entrenchment (BEBA) model. In their system, dynamic edge weights $w_{ij}(t) = 1 + \beta_i y_i(t) y_j(t)$ depend on cognitive entrenchment (β_i) and opinion alignment. Rather than forcing repulsion via an ad-hoc, artificial rule, they allowed negative influence ($w_{ij} < 0$) to emerge naturally from the consensus averaging equations. For a detailed critique of modeling bipolarization without explicit distancing, see Mäs & Flache (2013). We build directly on their mathematical foundation but transpose it into a pairwise, multi-issue interaction framework. We fuse their entrenchment formulation with Social Judgment Theory (Sherif & Hovland 1961; Jager & Amblard 2005), mapping weights to explicit zones of acceptance, indifference, and active repulsion, and overlaying a dynamic, co-evolving network of interpersonal trust.

Social Judgment Theory and the Backfire Effect

2.6 To pinpoint the thresholds where dialogue shifts from constructive assimilation to hostile repulsion, we deploy Social Judgment Theory (Sherif & Hovland 1961). This psychological theory divides an individual's attitude space into three distinct zones: the Latitude of Acceptance (agreeable concepts), the Latitude of Non-Commitment (neutral ground), and the Latitude of Rejection (highly objectionable ideas).

2.7 Exposure to a message in the Latitude of Acceptance (Sherif & Hovland 1961) prompts agents to assimilate the opinion, pulling their own views toward the sender. In stark contrast, messages landing in the Latitude of Rejection trigger a severe contrast effect: the receiver perceives the message as far more extreme than it actually is, causing them to push their own opinions away from the sender. This repulsion is the dreaded "backfire effect." Jager & Amblard (2005) integrated these dynamics into social networks, showing that active repulsion forces rapidly fracture uniform populations into stable, opposing camps. We map our dynamic influence weights directly onto these three psychological zones.

The Co-Evolution of Networks and Dyadic Trust

2.8 While early agent-based simulations treated network topologies as static backdrops, computational sociology and complex systems research demonstrates that opinions and social structures co-evolve dynamically (Macy & Willer 2002; Centola 2010; Keijzer et al. 2018). Most co-evolutionary models focus on structural plasticity, allowing agents to prune disagreeable edges and rewire their ties to similar neighbors. We chart a different path by focusing on "relational co-evolution": the network connections remain fixed, but their qualitative strength—mediated by dynamic, dyadic trust—co-evolves with dialogue.

2.9 Interpersonal trust mediates social influence. Risk and decision researchers have long highlighted a fundamental asymmetry in how humans trust: it is highly fragile—building slowly through repeated positive interactions, but eroding rapidly during conflict (Slovic 1993). In social networks, this means agreement slowly patches and strengthens dyadic ties, whereas backfire-inducing exchanges quickly corrode trust. Our model captures this fragility, introducing a crucial relational "buffer effect" (Slovic 1993): robust trust shields relationships from occasional ideological shocks, whereas collapsed trust turns even mild disagreements into triggers for active repulsion and decay.

Algorithmic Interventions and Exposure Experiments

2.10 The Polarization Paradox speaks directly to contemporary debates regarding algorithmic content moderation on digital platforms. Historically, platforms assumed that exposing users to cross-cutting, diverse viewpoints would naturally mitigate polarization. However, Bail et al. (2018) shattered this assumption in a landmark field experiment. They exposed Twitter users to a bot retweeting opposing political perspectives. Rather than cultivating open-mindedness, the intervention backfired: both Democrats and Republicans retreated to more extreme positions, demonstrating the real-world salience of opinion repulsion.

2.11 Subsequent empirical studies confirm that algorithmic feed design shapes political attitudes and user behaviors (Levy 2021; Guess et al. 2023). Platforms experiment with curation regimes that reduce exposure to divisive outgroup content; we abstract one such regime as a probabilistic "bridging" filter that intercepts rejection-zone interactions. Field studies typically examine short-term individual outcomes and cannot fully observe long-term systemic feedback among opinions, trust, and exit. Our agent-based model addresses that gap by tracing how suppression-oriented curation interacts with fragile dyadic trust over hundreds of conversational steps.

Model Description (ODD+D Protocol)

3.1 To guarantee absolute transparency and reproducibility, we formalize the Dynamic-BABE (Biased Assimilation & Behavioral Entrenchment) model via the standardized ODD+D (Overview, Design concepts, Details, and Decision-making) protocol (Grimm et al. 2010; Müller et al. 2013). This protocol provides a rigorous, structured architecture. It exposes the micro-level cognitive loops and decision heuristics that govern human interaction—the very forces driving online opinion dynamics.

Overview

Purpose

3.2 We built the Dynamic-BABE model to dissect the co-evolutionary feedback loop linking interpersonal trust, cognitive biases (such as biased assimilation and the backfire effect), and platform curation. By simulating these interlocking components, we lay bare the mechanics of the Polarization Paradox: superficial feed moderation arrests user exit and secures ad revenue today, only to entomb the social network in hostile ideological silos tomorrow.

Entities, State Variables, and Scales

3.3 Two entities drive the simulation: agents (users navigating their feeds) and the network environment (the network's structural pathways and the platform's moderation layer).

Agents:

3.4 Each agent $i \in \{1, 2, \dots, N_{\text{agents}}\}$ represents a single user, tracked via several state variables:

- Opinion Vector ($\vec{O}_i \in [-1, 1]^K$): A multi-dimensional ideological signature representing the agent's stance on K distinct social or political issues. The issue dimension $d \in \{1, \dots, K\}$ spans continuously from -1.0 (progressive/left) to $+1.0$ (conservative/right).

- Salience Vector ($\vec{S}_i \in [0, 1]^K$): An attention allocation vector mapping the relative cognitive importance the agent assigns to each issue. We constrain this vector so that salience sums to unity: $\sum_{d=1}^{K} S_{i,d} = 1.0$.

- Cognitive Entrenchment ($\beta_i \in [\beta_{\min}, \infty)$): A scalar tracking cognitive resistance, dogmatism, or confirmation bias (Chen et al. 2021). Elevated β_i values amplify outgroup rejection and fuel stronger backfire repulsion.

- Behavioral Frustration ($F_i \in \mathbb{N}_0$): An integer counter tracking the psychological irritation and cognitive dissonance that mount during confrontational, cross-cutting exchanges.

- Active Status ($\text{Active}_i \in \{\text{True}, \text{False}\}$): A boolean flag marking platform activity. When frustration breaches the critical threshold T_c, the agent permanently exits the platform ($\text{Active}_i = \text{False}$), registering as user churn.

Network and Platform Environment:

3.5 We model the social topology as an undirected, scale-free graph $G = (V, E)$ generated via the Barabási-Albert (BA) preferential attachment algorithm (Barabási & Albert 1999). The node set V represents agents, while the edge set E maps communication pathways.

- Dyadic Interpersonal Trust ($T_{ij} \in [0, 1]$): A symmetric weight representing relational capital on the edge $(i, j) \in E$. Trust cushions disagreement and co-evolves with opinion alignment. Disabling the trust dynamic locks T_{ij} at a static, neutral baseline $T_0 = 0.5$ across all edges.

- Algorithmic Interception Layer: The platform's computational gatekeeper. When active, it intercepts impending backfire events, neutralizing the psychological sting of deep ideological disagreement.

Scales:

3.6 Time ticks in discrete steps $t \in \{1, 2, \ldots, t_{\max}\}$, where each step represents a conversational epoch. Space is purely topological, dictated by network connections. The model populates a scale-free network with $N_{\text{agents}} = 500$ agents and preferential attachment parameter $m_e = 3$, yielding degree distributions that mirror real-world social networks.

Process Overview and Scheduling

3.7 Each step t triggers a sequence of operations that update opinions, frustration, trust, and churn:

1. Randomized Activation Order: The scheduler shuffles all active agents into a randomized list at the start of each step to eliminate sequential bias.

2. Agent Interaction Loop: We iterate through each active agent i in the shuffled list:

   (a) The agent filters its neighbors to isolate active peers: $\mathcal{N}_i^{\text{active}}(t) = \{j \in \mathcal{N}_i : \text{Active}_j(t) = \text{True}\}$. If this neighborhood is empty, the agent skips its turn.

   (b) The agent selects a dialogue partner j from $\mathcal{N}_i^{\text{active}}(t)$ uniformly at random. We set the interaction step size to $k_{\text{step}} = 1$ to capture micro-conversational dynamics.

   (c) The directed interaction pipeline $\text{Interact}(i, j)$ executes, updating agent i's opinion vector, accumulating frustration, checking for churn, and updating the symmetric trust T_{ij} on the connecting edge.

3. Passive Trust Decay: Once all active interactions conclude, we apply passive decay to every edge $(u, v) \in E$, including edges that were updated during the same step:

$$T_{uv}(t+1) = T_{uv}(t) \cdot (1 - \lambda) \tag{1}$$

4. Platform Diagnostics: The model computes and logs global metrics: opinion polarization $\sigma(t)$, cumulative churn, user frustration, active user count, ingroup/outgroup trust divergence, and opinion-weighted clustering.

5. Termination Check: The simulation halts when the step count reaches $t_{\max} = 200$, or if all users exit ($N_{\text{active}} = 0$).

Design Concepts

Theoretical Background

3.8 We anchor Dynamic-BABE's cognitive architecture in Social Judgment Theory (Sherif & Hovland 1961; Jager & Amblard 2005). This framework partitions an agent's attitude spectrum into three distinct zones: the Latitude of Acceptance (yielding assimilative agreement), the Latitude of Non-Commitment (resulting in inertia), and the Latitude of Rejection (triggering backfire repulsion).

3.9 To formalize cognitive entrenchment (β_i) and opinion alignment (A_{ij}), we adapt the Biased Assimilation and Behavioral Entrenchment (BEBA) model proposed by Chen et al. (2021), which extends DeGroot's classical consensus dynamics (DeGroot 1974). We merge this with a dynamic trust layer inspired by Slovic's empirical research (Slovic 1993). This integration captures the fundamental asymmetry of trust: relational capital accumulates painfully slowly through agreement but collapses instantly during conflict.

3.10 Finally, we operationalize user exit by mapping the accumulation of frustration to the classic Exit-Voice-Loyalty (EVL) framework of Hirschman (1970).

Individual Decision-Making

3.11 Agents employ heuristic cognitive updates rather than rational utility maximization, strictly following the bounded rationality principles of the ODD+D protocol.

- Objectives: Agents do not maximize an explicit utility or payoff function. Instead, their immediate psychological objective is to minimize cognitive dissonance. They seek agreeable social contact that reinforces their existing views while avoiding highly confrontational outgroup viewpoints that land in their Latitude of Rejection.
- Information Processing: When active, an agent perceives the opinion vector $\vec{O}_j$ and salience weights $\vec{S}_j$ of their chosen dialogue partner. They process this information heuristically by calculating their salience-weighted opinion alignment A_{ij} and checking whether the resulting influence weight lands in their Latitude of Acceptance, Non-Commitment, or Rejection.
- Adaptive Behavior: Agents adapt dynamically across two scales. Cognitively, they adjust their opinion vectors $\vec{O}_i$ to assimilate toward agreeable peers or repel away from hostile ones. Behaviorally, they accumulate frustration F_i when clashing with others, leading to permanent exit (churn) if their frustration threshold is breached. Relationally, their dyadic trust T_{ij} adapts to reward agreement and penalize conflict, reshaping future interaction weights.

Learning

3.12 Although agents lack formal reinforcement learning capabilities, they maintain a relational memory via dyadic trust (T_{ij}). They continuously update their assessment of neighbor compatibility. Agreeable, assimilative dialogue slowly accumulates trust, whereas hostile backfires instantly incinerate it. This co-evolving relational memory shapes future influence weights, closing the feedback loop between network relations and cognitive updates.

Individual Sensing

3.13 Each agent i tracks its direct topological neighbors with perfect accuracy. During dialogue, agent i observes a neighbor's opinion vector $\vec{O}_j$ and salience weights $\vec{S}_j$. Agents remain blind to the state variables, cognitive profiles, and degrees of non-neighboring nodes, and they hold zero global knowledge regarding system-level polarization or platform-level financials.

Individual Prediction

3.14 Agents react purely to immediate events, operating without forward-looking predictions. They neither anticipate neighbor opinion shifts nor forecast their own impending exit as frustration mounts toward the churn threshold.

Interaction

3.15 Interactions unfold locally, dyadically, and directionally. While trust updates symmetrically across the edge (i, j), opinion and frustration updates execute from the perspective of the active agent receiving the communication. The platform's bridging algorithm acts as an external filter, intercepting high-friction exchanges to suppress the behavioral symptoms of disagreement.

Collectives

3.16 No formal collectives (such as political parties or coalitions) exist in our model. However, functional collectives emerge endogenously through conversational dynamics. Agents coalesce into two opposing ideological factions based on their opinion signs. These emergent camps govern the flow of trust and drive the self-organization of network clusters.

Heterogeneity

3.17 We introduce substantial heterogeneity across several parameters:

- Ideological Camps: Bipolar initial opinions split the population into two opposed factions.
- Issue Salience: Diverse attention allocations drawn from a Dirichlet distribution mean agents prioritize issues differently.
- Cognitive Entrenchment: The entrenchment parameter β_i varies across agents, mixing dogmatic and open-minded individuals.
- Network Topology: The Barabási-Albert topology yields a heavy-tailed degree distribution where a few highly connected hubs exert massive social influence while the rest have minimal reach.

Stochasticity

3.18 We integrate stochasticity to capture social noise and scheduling dynamics:

- The preferential attachment model generates network topologies stochastically.
- Shuffling shatters sequential order by randomizing agent activation at each step.
- Dialogue partner selection is stochastic, choosing randomly from active neighbors.
- The bridging algorithm operates probabilistically, intercepting rejection events with an efficacy rate of $E = 46\%$.
- Initial values for opinion, salience, and entrenchment are drawn randomly from their respective probability distributions.

Observation

3.19 We record the system's trajectory at the end of each step by collecting system-level and agent-level diagnostics:

- Global Polarization ($\sigma(t)$): The root-mean-square (RMS) of per-dimension opinion standard deviations across active agents.
- Active User Count (N_{active}): The sum of active nodes.
- Churn Rate: The cumulative fraction of the population that has permanently exited.
- Average Frustration: The mean frustration across active users.
- Platform Revenue ($R(t)$): Total advertising earnings, which drop when polarization triggers advertiser flight.
- Trust Segregation Ratio: The ratio of mean ingroup trust to mean outgroup trust.
- Opinion-Weighted Clustering: The density of triadic relationships where all three connected nodes share the same opinion pole.

Details

3.20 This section details the mathematical formulations and structural rules governing initialization, dialogue, trust co-evolution, user exit, and platform monetization.

Initialization

3.21 At $t = 0$, the model builds an undirected, scale-free network G of $N_{\text{agents}} = 500$ nodes with attachment parameter $m_e = 3$.

3.22 To seed a pre-polarized society, we initialize opinion vectors $\vec{O}_i \in [-1, 1]^K$ ($K = 2$ issues) via a bipolar split. Each agent is assigned a camp pole $\pi_i \in \{-1, +1\}$ with equal probability. Independently for each issue dimension d, extremity is drawn from a uniform interval and scaled by the pole:

$$O_{i,d}(0) = \Pi_{[-1,1]}\Big(\pi_i \cdot U_{i,d}(0.4, 1.0)\Big) \tag{2}$$

where $U_{i,d}(0.4, 1.0)$ denotes an independent draw from $\mathcal{U}(0.4, 1.0)$ and $\Pi_{[-1,1]}$ clips opinions to $[-1, 1]$. The expected absolute extremity is 0.7, yielding two opposed, pre-polarized factions without additional Gaussian noise.

3.23 We draw each agent's salience vector $\vec{S}_i$ from a symmetric Dirichlet distribution:

$$\vec{S}_i \sim \text{Dirichlet}(\vec{\alpha}) \tag{3}$$

where $\vec{\alpha} = [1, 1]^T$ for $K = 2$ issues, ensuring that $S_{i,d} \in [0, 1]$ and $\sum_{d=1}^{K} S_{i,d} = 1.0$.

3.24 To model cognitive dogmatism, we draw agent entrenchment β_i from a normal distribution and floor it to prevent negative resistance:

$$\beta_i = \max\Big(\beta_{\min},\, \nu_i\Big), \quad \nu_i \sim \mathcal{N}(\mu_\beta,\, \sigma_\beta^2) \tag{4}$$

with parameters set to default values: $\beta_{\min} = 0.5$, $\mu_\beta = 4.0$, and $\sigma_\beta = 1.5$.

3.25 All active network edges $(u, v) \in E$ begin with symmetric dyadic trust set to a uniform starting value:

$$T_{uv}(0) = T_0 = 0.5 \tag{5}$$

Every agent begins the simulation with zero behavioral frustration ($F_i(0) = 0$).

Input Data

3.26 The Dynamic-BABE model is a theoretical, self-contained simulation and does not ingest external empirical input data for its execution.

Submodels

Dialogue Dynamics and Opinion Updates:

3.27 Upon activation, agent i randomly selects an active neighbor j for dialogue, processing the incoming opinion through a multi-stage cognitive pipeline.

Step 1: Salience-Weighted Opinion Alignment (A_{ij})

3.28 We calculate the ideological distance between receiver i and sender j as a salience-weighted dot product:

$$A_{ij} = \sum_{d=1}^{K} O_{i,d} \cdot \left(\frac{S_{i,d} + S_{j,d}}{2}\right) \cdot O_{j,d} \tag{6}$$

This formulation captures a key psychological reality: mutually important issues—where salience is high—contribute disproportionately to perceived agreement or disagreement. Because opinions are bounded in

$[-1, 1]$ and average saliences sum to 1.0, the alignment spans $A_{ij} \in [-1, 1]$, where $+1.0$ represents perfect alignment on highly valued topics and -1.0 marks absolute ideological clash.

Step 2: Perceived Influence Weight (w_{ij})

3.29 Adapting the BEBA model (Chen et al. 2021), we establish a baseline influence weight based on the receiver's cognitive entrenchment and opinion alignment:

$$w_{ij}^{\text{base}} = 1.0 + \beta_i A_{ij} \tag{7}$$

Under the Dyadic Trust system, relational trust modulates this weight, serving as a positive cognitive buffer:

$$w_{ij} = 1.0 + \beta_i A_{ij} + \gamma_T T_{ij} \tag{8}$$

where $\gamma_T \geq 0$ represents the trust influence coefficient ($\texttt{TRUST\_INFLUENCE} = 0.3$) and $T_{ij} \in [0, 1]$ represents the dyadic trust. This trust buffer mathematically models the relational buffer that keeps relationships intact during disagreement: high trust elevates the influence weight, preventing debate from deteriorating into hostile backfires. When trust co-evolution is disabled, the model sets $T_{ij} = 0$ dynamically in the weight equation, yielding the baseline form $w_{ij} = 1.0 + \beta_i A_{ij}$. This represents the unmodulated, baseline cognitive state and aligns the theoretical specification with our simulation codebase.

Step 3: Social Judgment Zone Classification

3.30 Unlike the collective, synchronous averaging of Chen et al. (Chen et al. 2021), we translate this cognitive logic into a local, pairwise interaction paradigm. To track individual metrics like frustration, churn, and algorithmic interventions, we classify interactions using Social Judgment Theory (Sherif & Hovland 1961; Jager & Amblard 2005). The perceived influence weight w_{ij} dictates the interaction zone:

$$\text{Zone}(w_{ij}) = \begin{cases} 1 & \text{(Latitude of Acceptance)} & \text{if } w_{ij} > \tau_{\text{assim}} \\ 2 & \text{(Latitude of Non-Commitment)} & \text{if } 0.0 \leq w_{ij} \leq \tau_{\text{assim}} \\ 3 & \text{(Latitude of Rejection)} & \text{if } w_{ij} < 0.0 \end{cases} \tag{9}$$

where $\tau_{\text{assim}} = 0.3$ represents the assimilation threshold. We model these latitude boundaries as stylized, uniform thresholds across the population for mathematical tractability and to isolate the systemic co-evolutionary feedback loops. In empirical social systems, these latitudes are highly subjective and dynamic, varying widely across individuals based on issue-specific emotional involvement and prior ideological extremity. We discuss the implications of this stylized assumption in Section .

Step 4: Algorithmic Moderation (The Bridge Algorithm)

3.1 When active, the platform's bridging algorithm intercepts backfire (Zone 3) events with an efficacy rate of $E = 0.46$:

$$w_{ij} \leftarrow 0.0, \quad \text{Zone}(w_{ij}) \leftarrow 2 \quad \text{with probability } E \tag{10}$$

Interception neutralizes the opinion repulsion, downgrading the conflict to Zone 2 (Ignore). In addition, successful moderation grants a frustration relief bonus to preserve the agent's psychological state. We model this frustration relief bonus ($B_{hb} = 2$) as the platform's active delivery of "dopaminergic comfort" or "algorithmic reinforcement" (e.g., stochastically substituting a hostile outgroup post with a soothing, highly engaging ingroup post or distracting entertainment). This intervention actively diverts attention, helping to soothe and heal preexisting cognitive friction.

Step 5: Opinion Update Rules

3.2 We update the opinion vector $\vec{O}_i$ depending on the final interaction zone:

- Zone 1 (Assimilation): The agent assimilates toward the sender using a DeGroot-style consensus update:

$$\vec{O}_i(t+1) = \Pi_{[-1,1]}\left(\vec{O}_i(t) + \mu_{ij}\left(\vec{O}_j(t) - \vec{O}_i(t)\right)\right) \tag{11}$$

where we attenuate the step size μ_{ij} by the agent's cognitive entrenchment:

$$\mu_{ij} = \frac{w_{ij}}{1.0 + \beta_i} \tag{12}$$

This ensures that highly entrenched agents (high β_i) make smaller opinion adjustments, reflecting cognitive resistance. The vector-wise operator $\Pi_{[-1,1]}$ acts element-wise to clip (clamp) updated opinions to the valid range $[-1.0, 1.0]$. We note that under conditions of absolute alignment ($A_{ij} = 1.0$) and high dyadic trust ($T_{ij} = 1.0$), the influence step size can stochastically exceed unity ($\mu_{ij} > 1.0$). While classical DeGroot consensus models enforce a strict convex combination ($\mu_{ij} \in [0, 1]$) to maintain the convex hull of opinions, our formulation permits this mathematical "overshoot." In the context of relational network dynamics, this represents a "passionate consensus" or "social reinforcement effect"—a psychological phenomenon where deep interpersonal trust and absolute agreement trigger cognitive enthusiasm, prompting the receiver to adopt a stance even more extreme than their trusted peer.

- Zone 2 (Non-Commitment): The interaction is ignored, leaving the opinion vector unchanged:

$$\vec{O}_i(t+1) = \vec{O}_i(t) \tag{13}$$

- Zone 3 (Backfire): The agent experiences ideological repulsion and pushes its opinion vector away from the sender:

$$\vec{O}_i(t+1) = \Pi_{[-1,1]}\left(\vec{O}_i(t) + \text{repulsion}_{ij}\left(\vec{O}_i(t) - \vec{O}_j(t)\right)\right) \tag{14}$$

where the repulsion step size is:

$$\text{repulsion}_{ij} = \frac{|w_{ij}|}{1.0 + \beta_i} \tag{15}$$

This repulsive force drives agent i further toward the ideological extremes, modeling the backfire effect, subject to the same element-wise clipping operator $\Pi_{[-1,1]}$.

Behavioral Frustration and User Churn:

3.3 Unmitigated disagreements accumulate psychological costs, which we model as behavioral frustration F_i:

- Frustration Accrual (Backfire): Unmitigated backfires (Zone 3) increment frustration:

$$F_i(t+1) = F_i(t) + 1 \tag{16}$$

- Frustration Healing (Assimilation): Assimilative dialogue (Zone 1) relieves cognitive dissonance, healing frustration at a constant rate $H = 1$:

$$F_i(t+1) = \max\left(0,\, F_i(t) - H\right) \tag{17}$$

- Moderation Relief Bonus: Successful algorithmic interception spares the agent the backfire and grants a mental relief bonus $B_{hb} = 2$:

$$F_i(t+1) = \max\left(0,\, F_i(t) - B_{hb}\right) \tag{18}$$

3.4 At the end of each interaction, the agent checks its psychological tolerance. When cumulative frustration breaches the churn threshold $T_c = 15$, the user permanently exits the platform ($\text{Active}_i = \text{False}$):

$$\text{Active}_i(t+1) = \begin{cases} \text{False} & \text{if } F_i(t+1) > T_c \\ \text{Active}_i(t) & \text{otherwise} \end{cases} \tag{19}$$

Churned users freeze instantly. They cease all dialogue, drop out of the communication graph, and generate zero advertising revenue.

Dyadic Trust Co-evolution:

3.5 Under the dynamic condition, dyadic trust T_{ij} co-evolves with dialogue outcomes:

$$T_{ij}(t+1) = \begin{cases} \min\Big(1.0,\, T_{ij}(t) + \delta_+\Big) & \text{if Zone}(w_{ij}) = 1 \text{ (Assimilation)} \\ \max\Big(0.0,\, T_{ij}(t) - \delta_-\Big) & \text{if Zone}(w_{ij}) = 3 \text{ (Backfire)} \\ T_{ij}(t) & \text{if Zone}(w_{ij}) = 2 \text{ (Ignore)} \end{cases} \tag{20}$$

where $\delta_+ = 0.02$ represents the slow accumulation of trust during agreement, and $\delta_- = 0.05$ captures the rapid collapse triggered by conflict. The resulting asymmetry ratio ($\delta_-/\delta_+ = 2.5$) mathematically formalizes Slovic's empirical trust asymmetry principle: social relationships are highly fragile, and conflict destroys trust much faster than cooperation builds it.

3.6 Because social ties naturally wither without active engagement, we apply a passive decay filter to every edge $(u, v) \in E$ at the end of each step:

$$T_{uv}(t+1) = T_{uv}(t) \cdot (1 - \lambda) \tag{21}$$

where $\lambda = 0.001$ represents the passive erosion rate.

Platform Revenue:

3.7 The social media platform monetizes active users, generating ad revenue at each step t. Advertisers flee volatile feeds, imposing a heavy Brand Safety Penalty if global polarization crosses the critical cliff:

$$R(t) = N_{\text{active}}(t) \cdot r_{\text{ad}} \cdot M_{\text{safety}}(t) \tag{22}$$

Here, $N_{\text{active}}(t)$ represents the active user count, $r_{\text{ad}} = \$1.00$ is the base ad rate per user, and $M_{\text{safety}}(t)$ represents the brand safety multiplier:

$$M_{\text{safety}}(t) = \begin{cases} M_{\text{safe}} = 1.0 & \text{if } \sigma(t) < \tau_{\text{cliff}} \\ M_{\text{unsafe}} = 0.4 & \text{if } \sigma(t) \geq \tau_{\text{cliff}} \end{cases} \tag{23}$$

where $\tau_{\text{cliff}} = 0.6$ represents the polarization cliff threshold. We calculate global polarization $\sigma(t)$ as the root-mean-square (RMS) of per-dimension standard deviations across active agents:

$$\sigma(t) = \sqrt{\frac{1}{K}\sum_{d=1}^{K} \text{SD}\Big(\{O_{i,d}(t) : i \in \mathcal{A}(t)\}\Big)^2} \tag{24}$$

where $\mathcal{A}(t) = \{i \in V : \text{Active}_i(t) = \text{True}\}$ and SD denotes the population standard deviation within dimension d. This formulation avoids the "data flattening" trap: averaging opinions across dimensions can cancel opposed extremes (e.g., $[+1, -1]$), whereas per-dimension dispersion preserved under RMS does not.

3.8 Ingroup and outgroup trust metrics classify an agent's pole by $\text{sign}(\overline{O}_i)$, the sign of the mean opinion across issues. Agents with mean opinion exactly zero ($\text{sign}(0) = 0$) are rare under bipolar initialization but, when present, are treated as a distinct zero pole for edge classification; we discuss this edge case as a limitation.

Experimental Design

3.9 We map the co-evolutionary dynamics above with a 2×2 factorial design crossing the Bridging Algorithm and Dyadic Trust co-evolution:

1. Baseline (Bridge OFF, Trust OFF): Biased assimilation and entrenchment under static neutral trust ($T_{ij} = 0.5$).
2. Bridge Only (Bridge ON, Trust OFF): Bridging active; trust static.
3. Trust Only (Bridge OFF, Trust ON): Trust co-evolves; no bridging.
4. Full Model (Bridge ON, Trust ON): Bridging and trust co-evolve together.

Table 1: Model Parameters and Default Values

Parameter Symbol	Description	Default Value	Source / Rationale
N_{agents}	Population size	500	Config.py standard pop. size
m_e	BA network attachment parameter	3	Scale-free social degree distribution
K	Number of opinion dimensions	2	Multidimensional issue space
β_{mean}	Mean of cognitive entrenchment	4.0	High entrenchment social media bias
β_{std}	SD of cognitive entrenchment	1.5	Cognitive heterogeneity in population
β_{min}	Minimum entrenchment floor	0.5	Prevent negative cognitive resistance
k_{step}	Dialogue partners selected per step	1	Micro-interaction conversational model
τ_{assim}	Assimilation threshold	0.3	Social Judgment Theory threshold
T_c	Churn frustration threshold	15	Toxic churn tolerance capacity
H	Natural frustration healing rate	1	Cognitive cooling per positive contact
B_{hb}	Bridge moderation healing bonus	2	Active relief under algorithmic safety
r_{ad}	Base advertising rate per step	\$1.00	Baseline monetization rate
τ_{cliff}	Brand safety polarization cliff	0.6	Advertiser flight tipping point
M_{safe}	Brand safe revenue multiplier	1.0	Normal ad monetization
M_{unsafe}	Brand unsafe revenue multiplier	0.4	60% discount due to toxic environment
E	Bridging algorithm efficacy	0.46	46% moderation success probability
T_0	Initial dyadic trust weight	0.5	Relational neutrality starting state
γ_T	Trust buffer influence coefficient	0.3	Disagreement buffering coefficient
δ_+	Trust building step size (gain)	0.02	Slow relational capital accumulation
δ_-	Trust erosion step size (loss)	0.05	Slovic relational fragility parameter
λ	Passive trust decay rate	0.001	Under-maintenance erosion rate
t_{max}	Maximum simulation step limit	200	Conversational equilibrium epoch

Model Hyperparameters and Topology

3.10 The model embeds $N = 500$ agents in a Barabási–Albert network (Barabási & Albert 1999) with attachment parameter $m = 3$. Opinions occupy $K = 2$ issues under bipolar initialization: each agent receives pole $\pi_i \in \{-1, +1\}$ with equal probability, and each issue is set to $\pi_i \cdot U(0.4, 1.0)$ (clipped to $[-1, 1]$), so expected absolute extremity is 0.7.

3.11 Cognitive resistance β_i is drawn $\mathcal{N}(\mu_\beta = 4.0, \sigma_\beta = 1.5)$ with floor $\beta_{\text{min}} = 0.5$. When enabled, trust shifts influence by $\gamma_T = 0.3$, gains $\delta_+ = 0.02$ on assimilation, loses $\delta_- = 0.05$ on backfire, and decays at $\lambda = 0.001$ per step on all edges after interactions. Bridging intercepts Latitude-of-Rejection events with efficacy $E = 46\%$ and grants a two-point frustration relief bonus.

3.12 We run $R = 30$ replications per condition for $S = 200$ steps. Reproducibility uses base seed 42; run i uses seed $42 + i$, so network structure and stochastic draws are reconstructible for each replicate.

Statistical Methodology

3.13 We analyze final-step metrics across all 120 runs with non-parametric methods. Pairwise condition contrasts use two-sided Mann–Whitney U tests (Mann & Whitney 1947) with Bonferroni correction across the full pairwise family, and Cohen's d (Cohen 1988) for standardized magnitude (negligible $|d| < 0.2$; small < 0.5; medium < 0.8; large ≥ 0.8). We also report 95% bootstrap confidence intervals (10,000 resamples).

3.14 For Bridge × Trust moderation, we compute seed-paired effects $\Delta_{\text{alone}} = (\text{Bridge Only}) - (\text{Baseline})$ and $\Delta_{\text{with trust}} = (\text{Full Model}) - (\text{Trust Only})$, then apply a Wilcoxon signed-rank test to $\Delta_{\text{with trust}} - \Delta_{\text{alone}}$, with Bonferroni correction across interaction metrics. This paired procedure matches the shared-seed design more closely than an unpaired rank-sum test on the two effect vectors.

Results

4.1 Our 2×2 factorial simulation experiments yield clear quantitative patterns. By crossing the platform's Bridging Algorithm (ON vs. OFF) with the co-evolutionary Dyadic Trust framework (ON vs. OFF), we run 30 independent simulations per condition—each running for 200 steps. Reproducibility uses base seed 42, with run i initialized at seed $42 + i$. To compare these states, we employ two-sided, non-parametric

Mann-Whitney U tests backed by Bonferroni corrections, and compute Cohen's d to quantify standardized effect sizes (Cohen 1988). Large Cohen's d values for churn and revenue reflect nearly non-overlapping condition distributions rather than subtle mean shifts; we therefore emphasize absolute mean differences alongside standardized effects.

The Polarization Paradox: Short-Term Monetization vs. Long-Term Alignment

4.2 We begin by comparing the Baseline and Bridge Only conditions, isolating the impact of algorithmic feed moderation when network trust remains static. The simulations demonstrate a tension between a platform's short-term financial retention and long-term ideological alignment—a structural deadlock we term the Polarization Paradox.

User Retention and Platform Revenue

4.3 In the unmoderated Baseline, cumulative churn reaches 39.18% (SD = 2.27%, 95% CI = $[38.39\%, 39.98\%]$) by step 200, leaving only 304.1 active users on average (SD = 11.35, 95% CI = $[300.10, 308.03]$). Frustration among the survivors rises to 0.9668 (SD = 0.1669, 95% CI = $[0.9055, 1.0239]$). This user flight depresses platform economics: final step-wise revenue falls to a mean of \$121.64 (SD = \$4.54, 95% CI = $[\$120.04, \$123.21]$).

4.4 Activating the bridging algorithm (Bridge Only condition) changes retention sharply. Cumulative churn falls to 0.38% (SD = 0.30%, 95% CI = $[0.27\%, 0.49\%]$) at step 200 (see Figure 1). The contrast is highly significant: Mann-Whitney $U = 900.0$, $p_{\text{adjusted}} < 1.5 \times 10^{-9}$, Cohen's $d = 23.97$. The platform retains 498.1 active users (SD = 1.52, 95% CI = $[497.57, 498.63]$)—nearly the full population.

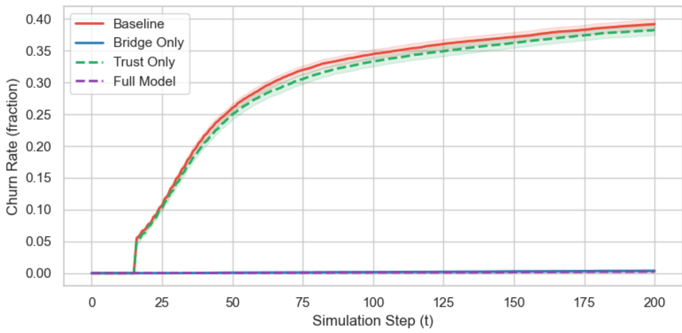

Figure 1: Cumulative user churn rate over time across the four experimental conditions ($N = 500$, 30 runs per condition). The Bridge ON conditions show near-zero user flight (0.38%), while the Bridge OFF conditions display high user attrition ($\approx 39\%$). Error ribbons indicate 95% confidence intervals.

4.5 Algorithmic shielding lowers mean frustration among active users to 0.5819 (SD = 0.0576, 95% CI = $[0.5617, 0.6024]$, Mann-Whitney $U = 871.0$, $p_{\text{adjusted}} < 3.1 \times 10^{-8}$, Cohen's $d = 3.08$). Retaining users stabilizes final step-wise revenue at \$199.24 (SD = \$0.61, 95% CI = $[\$199.03, \$199.45]$, Mann-Whitney $U = 0.0$, $p_{\text{adjusted}} < 1.5 \times 10^{-9}$, Cohen's $d = -23.97$). This approximately 64% revenue increase is driven by retention of active users rather than improved feed safety. Because the initial society is pre-polarized ($\sigma(0) \approx 0.7 > 0.6$), the stylized brand-safety multiplier remains in the unsafe regime ($M_{\text{safety}} = 0.4$) for both conditions. The financial benefit is therefore demographic: the algorithm keeps users active and preserves ad impressions even when content remains polarized.

The Paradoxical Ideological Backlash

4.6 Beneath the financial surface, filtering confrontational interactions yields a statistically significant increase in global ideological polarization. Baseline polarization σ settles at a mean of 0.9550 (SD = 0.0389, 95% CI = $[0.9409, 0.9683]$), while Bridge Only raises final-step polarization to 0.9904 (SD = 0.0042, 95% CI = $[0.9888, 0.9918]$)—a significant increase (Mann-Whitney $U = 135.0$, $p_{\text{adjusted}} < 0.0002$, Cohen's $d = -1.28$; see Figure 2).

4.7 A structural selection effect explains this phenomenon. In the unmoderated Baseline, extreme, dogmatic agents repeatedly clash with their opponents. These interactions trigger backfires, pile up frustration,

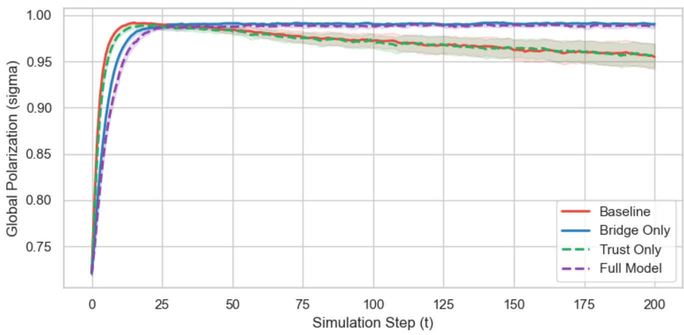

Figure 2: Global opinion polarization (σ) over time across the four experimental conditions. While the Bridging Algorithm suppresses churn in the Bridge Only condition, it paradoxically locks systemic polarization at a significantly higher level ($\sigma \approx 0.9904$) compared to the Baseline ($\sigma \approx 0.9550$), demonstrating the Polarization Paradox. Error ribbons represent 95% confidence intervals.

and drive agents to churn. Their permanent exit mathematically clips the tails of the opinion distribution, capping global polarization.

4.8 By contrast, the bridging algorithm isolates extreme users from frustration and keeps them active. Retained ideologues continue to pull neighbors toward the extremes, reducing the chance of consensus. By mitigating the symptoms of polarization—exit and frustration—the algorithm can preserve high systemic dispersion, locking the system in ideological deadlock.

Trust Segregation: The Relational Cost of Social Judgments

4.9 Enabling dyadic trust (Trust Only condition) triggers a dramatic structural transformation. Trust co-evolves with dialogue as a dynamic link attribute, reshaping the network's qualitative architecture.

4.10 In our Baseline, trust remains locked at a static nominal value of 0.5 across all edges, presenting a flat relational landscape where ingroup and outgroup trust are identical (yielding a segregation ratio of exactly 1.00).

4.11 Under the Trust Only condition, co-evolving opinions and trust split the network into relational camps (see Figure 3). Ingroup trust—the trust linking agents sharing the same opinion pole—climbs to a near-perfect mean of 0.9814 (SD $= 0.0055$, 95% CI $= [0.9794, 0.9832]$). Conversely, outgroup trust completely collapses, bottoming out at a mean of 0.0415 (SD $= 0.0256$, 95% CI $= [0.0331, 0.0511]$).

4.12 This relational divergence raises the Trust Segregation Ratio to a mean of 35.61 (SD $= 31.50$, 95% CI $= [26.29, 47.64]$). Because the ratio is heavy-tailed (maximum ≈ 175 across runs), we also report the median and interquartile range: median $= 28.58$, IQR $= [19.06, 40.61]$. Compared to the Baseline, the effect is large: Mann-Whitney $U = 0.0$, $p_{\text{adjusted}} < 7.3 \times 10^{-11}$, Cohen's $d = -1.55$. These figures describe the Trust Only vs. Baseline contrast (enabling trust co-evolution), not a bridging effect.

4.13 This segregation exposes the double-edged role of trust. Inside ideological camps, high ingroup trust increases the influence weight w_{ij}, buffering minor disagreements via a "friends can argue" mechanism. Across camps, the asymmetry of trust co-evolution ($\delta_-/\delta_+ = 2.5$) erodes cross-cutting ties faster than agreement can rebuild them. Combined with passive decay ($\lambda = 0.001$), outgroup trust collapses and the network segregates into two cohesive but mutually hostile relational silos.

Moderation Analysis: The Bridge × Trust Interaction

4.14 To test whether co-evolving dyadic trust moderates the impact of algorithmic content moderation, we perform a seed-paired interaction analysis. Because runs share random seeds, we compute paired differences in final metrics:

$$\Delta_{\text{alone}} = \text{Metric(Bridge Only)} - \text{Metric(Baseline)} \tag{25}$$

$$\Delta_{\text{with trust}} = \text{Metric(Full Model)} - \text{Metric(Trust Only)} \tag{26}$$

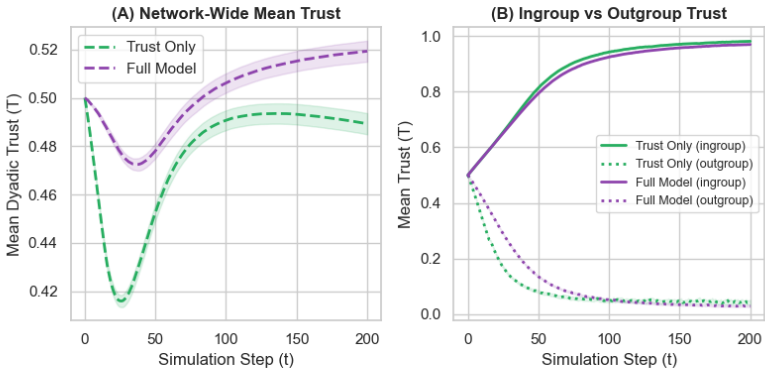

Figure 3: Interpersonal dyadic trust co-evolution dynamics. Panel A compares mean trust over time in the Trust Only and Full Model. Panel B illustrates the extreme divergence between ingroup trust (climbing toward 0.98) and outgroup trust (collapsing to near 0.03), demonstrating the emergence of severe trust segregation. Error ribbons denote 95% confidence intervals.

Table 2: Bridge × Trust Interaction Analysis (Seed-Paired Differences)

Metric	Effect Alone (Δ_{alone})	Effect w/Trust ($\Delta_{\text{with trust}}$)	Interaction Cohen's d	W-statistic	Adj. p-value
Active Users	+194.0	+190.1	−0.3508	112.5	0.1346
Avg. Frustration	−0.3849	−0.3981	−0.0739	198.0	1.0000
Churn Rate	−38.80%	−38.02%	+0.3508	116.5	0.1698
Ingroup Trust	0.0000	−0.0114	−2.6925	4.0	$< 1.4 \times 10^{-7}$
Mean Trust	0.0000	+0.0300	+4.8926	0.0	$< 1.9 \times 10^{-8}$
Outgroup Trust	0.0000	−0.0127	−0.8219	97.0	0.0434
Polarization	+0.0353	+0.0322	−0.0887	187.0	1.0000
Revenue	+$77.60	+$76.04	−0.3508	106.5	0.0954
Trust Segregation	0.0000	+1.9854	+0.0902	135.0	0.4491

We then apply a Wilcoxon signed-rank test to the per-seed contrast $\Delta_{\text{with trust}} - \Delta_{\text{alone}}$, which matches the paired design (see Table 2).

Active Users, Frustration, and Churn:

4.15 Interaction effects on physical user retention (Active Users, Churn Rate, and Avg. Frustration) and financial performance (Revenue) remain non-significant after Bonferroni correction in the seed-paired analysis. The bridging algorithm's retention benefits are similar with or without trust co-evolution: it rescues approximately 190 to 194 users from churning, reduces frustration by ≈ 0.39, and preserves $\approx \$76$ in step-wise revenue.

Outgroup Trust and Trust Segregation:

4.16 Relational metrics show a nuanced pattern. Seed-paired Wilcoxon tests find a significant Bridge $\times$ Trust interaction for Outgroup Trust ($W = 97.0$, $p_{\text{adjusted}} = 0.0434$, interaction $d = -0.82$): when trust co-evolves, the bridge effect on outgroup trust is a small mean reduction of -0.0127 (Full Model 0.0288 vs. Trust Only 0.0415). By contrast, the Trust Segregation interaction is not significant after Bonferroni correction ($W = 135.0$, $p_{\text{adjusted}} = 0.4491$, $d \approx 0.09$). Pairwise Full Model vs. Trust Only contrasts for both Outgroup Trust and Trust Segregation are also non-significant after Bonferroni correction ($p_{\text{adjusted}} = 1.000$).

4.17 We therefore treat bridging's relational side-effects as modest and method-dependent: retention benefits are robust, while claims of intensified segregation under the Full Model are not supported by the paired Wilcoxon interaction test or by pairwise condition contrasts. Mechanistically, bridging can replace rejection-zone events with Zone 2 silence (no trust rebuild) and retain agents who would otherwise churn, leaving ties exposed to passive decay ($\lambda = 0.001$); these mechanisms can shift descriptive means without producing a large, reliably detected segregation interaction.

Echo Chambers: Emergent Structural Clustering

4.18 To analyze how these cognitive and relational forces reshape the topological communication patterns of the platform, we track the Opinion-Weighted Clustering Coefficient (see Figure 4). This metric measures the local density of triadic relationships where all three connected agents share the same opinion pole—serving as a direct indicator of structural echo chambers.

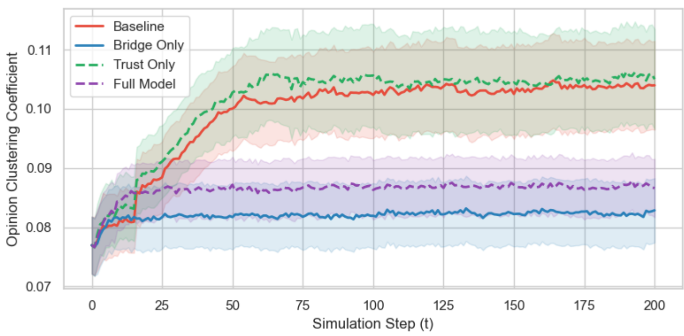

Figure 4: Emergent structural echo chambers, measured via the opinion-weighted clustering coefficient over time. The Bridging Algorithm (Bridge Only and Full Model) significantly reduces local triadic homogeneity compared to the Baseline, although global polarization remains extreme. Error ribbons show 95% confidence intervals.

4.19 In our Baseline, the final mean opinion-weighted clustering coefficient settles at 0.1040 (SD = 0.0209, 95% CI = $[0.0968, 0.1115]$). When we activate trust co-evolution alone (Trust Only), clustering remains statistically unchanged at a mean of 0.1050 (SD = 0.0237, 95% CI = $[0.0968, 0.1134]$, Mann-Whitney $U = 420.0$, $p_{\text{adjusted}} = 1.0000$, Cohen's $d = -0.05$). This static result shows that trust co-evolution alone, despite locking relationships into hostile camps, does not alter the physical clustering of opinions beyond the baseline scale-free structure.

4.20 Deploying the bridging algorithm reduces structural clustering. Under the Bridge Only condition, opinion-weighted clustering drops to a mean of 0.0829 (SD = 0.0152, 95% CI = $[0.0774, 0.0881]$)—a significant decline relative to the Baseline (Mann-Whitney $U = 721.0$, $p_{\text{adjusted}} = 0.0038$, Cohen's $d = 1.16$).

4.21 Similarly, under the Full Model (Bridge ON, Trust ON), opinion-weighted clustering settles at a mean of 0.0866 (SD = 0.0137, 95% CI = $[0.0816, 0.0914]$), also significantly below Baseline (Mann-Whitney $U = 681.0$, $p_{\text{adjusted}} = 0.0393$, Cohen's $d = 0.98$).

4.22 This shift occurs because bridging intercepts many extreme backfires and dampens local repulsion. Global polarization can remain high because dogmatic users who would otherwise exit are retained, while local triangles become slightly less homogeneous.

4.23 Lower clustering should not be read as systemic resolution. Agents may remain connected in triangles while interpersonal trust stays partitioned and many cross-cutting channels are inactive.

Discussion

5.1 Our experiments describe a co-evolutionary system in which algorithmic filters, cognitive biases, and dyadic relationships interact. Here we interpret the Polarization Paradox through Exit–Voice–Loyalty, discuss the double-edged role of trust, outline design implications, report a brief robustness check, and note limitations.

The Polarization Paradox: An Exit-Voice-Loyalty Analysis

5.2 Hirschman (1970)'s Exit, Voice, and Loyalty framework clarifies the paradox. When quality decays—here, through rising ideological conflict—members may Exit (leave) or exercise Voice (debate and repair).

5.3 In the unmoderated Baseline, entrenched agents encounter outgroup opinions that land in the Latitude of Rejection, triggering backfires and frustration. Lacking an effective repair channel, many choose Exit and churn. That exit is costly for the platform, but it also removes radical agents and can clip the tails of the opinion distribution.

5.4 Bridging targets the economic threat by intercepting many backfires, closing the Exit pathway for frustrated users. Retention and revenue improve. The intervention does not, however, cultivate Voice: rejection-zone events are often replaced by Zone 2 silence rather than constructive engagement. This containment is not Hirschmanian Loyalty; it is synthetic retention without a parallel channel for repair. Closing exit while blocking voice can trap extreme agents inside the network, sustaining high polarization even as churn falls.

The Double-Edged Sword of Dyadic Trust

5.5 Dynamic trust cuts both ways. Like-minded agents accumulate relational capital ($\delta_+ = 0.02$), raising mean ingroup trust to 0.9814 and buffering within-camp disagreement. Across camps, asymmetry ($\delta_-/\delta_+ = 2.5$) following Slovic (1993) depletes outgroup trust to a mean of 0.0415 in the Trust Only condition. With passive decay ($\lambda = 0.001$), the network segregates (mean Trust Segregation Ratio 35.61; median 28.58). These relational outcomes are primarily attributable to enabling trust co-evolution, not to the bridging contrast that defines the Polarization Paradox.

Platform Design and Architectural Implications

5.6 Many curation systems optimize engagement, retention, and brand safety. Suppression can treat behavioral symptoms (outrage, exit) while leaving entrenchment and fragile trust intact. Our results suggest that retention gains need not imply cohesion gains. Multi-objective designs that also track cross-camp relational capital—for example, connecting users via shared non-political salience before exposing political disagreement—offer a constructive alternative to pure avoidance. These implications are normative design hypotheses, not empirical prescriptions for any specific platform.

Robustness to Parameter Variation

5.7 We conducted one-factor-at-a-time (OFAT) sensitivity sweeps over β_{mean}, τ_{assim}, T_c, K, γ_T, E, and m_e (five replications per cell; $N = 400$ runs; see `sensitivity.py`, `output/sensitivity_results.csv`, and the generated sensitivity figures). Across nearly all interior sweep points, the qualitative Polarization Paradox pattern holds: bridging sharply reduces churn while failing to reduce—and typically elevating—final polarization. Absolute levels of churn and polarization shift with parameters (especially β_{mean}, T_c, E, and m_e), but directional bridge-versus-baseline contrasts reverse only at identifiable boundaries. At very low entrenchment ($\beta_{\text{mean}} \in \{1, 2\}$), both conditions remain near-consensus with negligible churn, so the paradox is undefined rather than contradicted. Setting bridge efficacy to $E = 0$ yields a null intervention (churn essentially unchanged). At the opposite pole, $E = 1.0$ retains the churn reduction but collapses polarization—suggesting that only near-perfect suppression, not the moderate bridging used in the main design, dissolves the paradox on the opinion axis. Full numerical tables and per-parameter figures are available in the replication repository.

Limitations and Future Work

5.8 Although Dynamic-BABE provides a transparent theoretical laboratory, several boundaries remain:

1. Symmetric Trust: We assume $T_{ij} = T_{ji}$. Directed, asymmetric trust is an important extension.
2. Static Undirected Topology: Barabási–Albert graphs are undirected and fixed. Real platforms often use directed follower ties with higher clustering and ongoing rewiring. Absolute clustering levels should be read as stylized signals. Directed hubs that broadcast without reciprocal friction may alter polarization and churn.
3. Low-Dimensional Issue Space: We use $K = 2$. Higher-dimensional issue spaces and cross-cutting cleavages may change global dynamics.
4. Pole Classification Edge Case: Ingroup/outgroup metrics use $\text{sign}(\overline{O}_i)$. Agents with mean opinion exactly zero form a distinct zero pole; such cases are uncommon under bipolar initialization but remain a modeling limitation.
5. Empirical Calibration: Parameters are literature-motivated, not fitted to a named platform dataset. Field calibration remains future work.
6. Stylized Bridging and Revenue: The bridge filter and brand-safety multiplier are abstractions for theoretical clarity, not audited replicas of production ranking or ad systems.

5.9 These limits notwithstanding, the framework offers a reproducible basis for studying how suppression and fragile trust jointly shape polarization and retention.

Conclusion

6.1 We mapped co-evolutionary loops linking opinion dynamics and relational networks under algorithmic suppression and fragile interpersonal trust. With Dynamic-BABE, we identify a Polarization Paradox: bridging-style interventions can sharply improve retention and step-wise revenue while raising final polarization by retaining extreme agents who would otherwise exit. Separately, dyadic trust co-evolution produces trust segregation—high ingroup trust and collapsed outgroup trust—even without bridging. Seed-paired Bridge × Trust analyses indicate that retention benefits are robust to trust dynamics, whereas relational moderation effects are modest and should be interpreted cautiously when pairwise contrasts do not survive multiple-comparison correction.

6.2 These results argue for multi-objective platform design that treats relational capital as a first-class outcome alongside engagement and brand safety. Spaces that seed cross-camp trust on shared non-political saliences may convert hostile outgroup contact into constructive voice. Escaping the Polarization Paradox requires more than suppressing conflict; it requires sustaining the contact conditions under which tolerance can form.

Code and Data Availability

6.3 The Dynamic-BABE implementation, batch scripts, analysis pipeline, and figure generators are publicly available at `https://github.com/TheCoolSam/BABEModel`. We intend to deposit a reviewed release on the CoMSES Net Computational Model Library so that a permanent handle can accompany the published article.

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Notes

[1]CoMSES upload to be completed by the authors prior to publication.

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